

Scope of Work: Q1 Research Paper

Reinforcement Learning for Dynamic Project Portfolio Budgeting under Cashflow Uncertainty: A Rolling Horizon Approach

1. Research Problem

1.1 Core Challenge

Project portfolio budgeting under cashflow uncertainty requires dynamic decision-making in environments where:

- Future contractor performance is unpredictable
- Traditional optimization methods fail due to incomplete observability
- Sequential decisions must adapt to emerging information
- Projects have staggered start times and variable durations
- Portfolio horizons extend beyond fixed planning windows

1.2 Limitations of Existing Approaches

Classic Operations Research (OR) algorithms (MIP, Stochastic Programming) are inadequate because:

- They require complete observability of the environment
- They assume known probability distributions of future uncertainties
- They cannot adapt to real-time information updates
- They fail to handle the sequential nature of budget allocation decisions
- Scenario tree formulations become computationally intractable for variable-horizon portfolios

1.3 Research Gap

As of 2026, no practical, validated models exist for portfolio budgeting optimization that:

- Handle cashflow uncertainty dynamically
- Adapt to company-specific historical patterns
- Provide sequential decision-making capabilities under partial observability
- Accommodate variable-length portfolios with staggered project lifecycles
- Balance computational tractability with real-world complexity

2. Proposed Solution

$P(s_{t+1}|s_t, a_t)$ is unknown due to contractor performance uncertainty

Traditional methods require:

- Known transition probabilities → **unavailable**
- Complete scenario enumeration → **intractable**
- Static optimization → **incompatible with sequential decisions**

RL provides:

- Model-free learning from experience
- Adaptive policy under uncertainty
- Scalable to real-time decision-making

2.1 Methodology

Train a Reinforcement Learning (RL) agent using **Rolling Horizon Control** with:

1. **Pre-training**: Literature and industry-wide project data
2. **Fine-tuning**: Company-specific historical performance data
3. **Rolling Horizon Framework**: Fixed lookahead window ($H=12$ periods) with receding control

2.2 Rolling Horizon Architecture

Core Concept

At each timestep t , the system:

1. Observes current portfolio state (active projects, budget status, performance trends)
2. Plans budget allocation for next $H = 12$ periods
3. Executes only the allocation for period t

4. Advances to $t + 1$ and re-plans with updated information

Key Properties

- **Fixed Episode Length:** RL trains on 12-period episodes regardless of total portfolio duration
- **Variable Portfolio Horizons:** Naturally handles portfolios spanning 12, 24, 36+ periods through sequential re-planning
- **Dynamic Project Management:** Accommodates projects starting/finishing at different times through masking and padding mechanisms
- **Computational Tractability:** Maintains feasible state-space size for both RL and OR baselines

Handling Variable Project Lifecycles

The rolling horizon framework uses **masking and padding** to handle:

- Projects that start after the current timestep (masked until start date)
- Projects that complete before the horizon end (masked after completion)
- Projects with partial completion at portfolio initialization
- Variable project durations and staggered entry/exit times

This mechanism allows the model to handle arbitrary project configurations without requiring all projects to be new at $t = 0$.

2.3 Key Advantages of RL Approach

- **Adaptability:** Learns from sequential interactions rather than requiring complete upfront knowledge
- **Partial Observability:** Handles uncertainty without needing full probability distributions
- **Dynamic Optimization:** Adjusts decisions as new information emerges through rolling re-planning
- **Transferability:** Pre-trained model can be specialized for specific organizational contexts
- **Scalability:** Rolling horizon prevents state-space explosion for long-duration portfolios
- **Real-World Alignment:** Mimics actual planning behavior (annual budgets with periodic updates)

3. Assumptions

3.1 Industry Context: EPC Oil & Gas Projects

- **The model is calibrated for Engineering, Procurement, and Construction (EPC) contractors in the oil & gas sector**
- Project cashflow profiles reflect typical EPC spending patterns observed in literature
- S-curve parameters (shape, peak timing, skewness) are derived from empirical studies of large-scale energy infrastructure projects

Justification:

- EPC projects exhibit relatively predictable cashflow structures due to standardized contracting practices
- Oil & gas sector provides rich historical data for S-curve calibration
- Findings remain generalizable to other capital-intensive project portfolios with similar contractual frameworks

3.2 Foundational Principle: Simplest Model

Rationale: This is a brand new field in the literature. The Q1 paper establishes foundational proof-of-concept.

Strategy: Extension of each assumption will be explicitly recognized as future work.

3.3 Project Structure Assumptions

Cashflow-Based Budgeting Assumption

- **All budgeting decisions are made solely using aggregated project cashflows (inflows and outflows)**
- Detailed resource-level requirements (labor, materials, equipment) are **not modeled individually**
- Project execution schedules and resource demands are **compressed into a contractual S-curve spending profile**
- Portfolio optimization operates on these S-curves rather than task-level or resource-level data

Justification:

- S-curve budgeting is the industry standard for high-level financial planning
- Avoids unnecessary granularity inconsistent with Q1 scope

- Ensures tractable state and action spaces for RL models
- Maintains focus on strategic financial decisions rather than operational scheduling

3.4 Payment & Performance Method

- **All projects use Earned Value Management (EVM)** for performance measurement and payment
- Payments are tied to earned value milestones
- Contractor performance is measured through EVM metrics

Justification:

- EVM is industry-standard for project control
- Provides consistent performance measurement framework
- Enables quantifiable uncertainty modeling

3.5 Aggregate Performance Uncertainty

- **All project uncertainties are modeled as a single aggregated performance risk factor**
- Human resource variability, inflation effects, supply chain disruptions, and regulatory impacts are **not treated separately**
- Uncertainty enters the model through a unified performance noise term affecting project progress and cost
- No decomposition of uncertainty sources; only portfolio-level aggregate volatility is modeled

Justification:

- Simplifies stochastic modeling and avoids multi-factor calibration
- Keeps the environment stationary and tractable for RL/optimization
- Enables consistent uncertainty treatment across heterogeneous projects
- Prevents overfitting to specific external factors that are outside Q1 scope

Uniform S-Curve Shape Across All Projects

- **All projects follow the same Beta CDF parameterization** with $\alpha = 2.5$, $\beta = 2.0$ for cost progression
- **No project-specific S-curve variations** based on project type, size, or complexity
- **Peak spending occurs uniformly at ~43% completion** for all projects in the portfolio
- **Front-loading intensity is constant** regardless of engineering/procurement/construction phase mix

Justification:

- Prevents state space explosion from adding $2N$ shape parameters (α_i, β_i per project)
- Maintains consistency with foundational simplicity principle (Section 3.2)
- Literature shows EPC projects cluster around similar spending profiles (Barraza & Bueno 2007)
- Portfolio-level aggregation smooths individual project variations
- Strategic budget allocation decisions depend on aggregate cashflow dynamics, not micro-level spending patterns
- Sensitivity analysis can demonstrate policy robustness across reasonable parameter ranges

Fixed 10% Profit Margin (Cost-Plus Pricing)

- **All projects priced at Contract Value = BAC $\times$ 1.10** regardless of risk, complexity, or client
- **Profit margin is deterministic and non-negotiable** (no competitive bidding dynamics)
- **ROI target is uniform across portfolio** and does not vary by project characteristics

Justification:

- Reflects government value-added regulations mandating 10% ROI for public sector contracts
- Eliminates pricing uncertainty to focus on budget allocation and liquidity management under payment delays
- Consistent with Cost-Plus contracting norms in oil & gas EPC sector (Section 3.1)
- Simplifies revenue modeling while maintaining realistic profit dynamics
- Aligns with foundational simplicity principle: single contractual framework across portfolio

Revenue Recognition Follows Cost Progress (Percentage-of-Completion with Stochastic EV)

- **Revenue S-curve has identical shape to cost S-curve** (same α, β parameters) under baseline planned conditions
- **Percentage-of-Completion method ties revenue recognition to Earned Value (EV)**, not actual cost incurred
- **Earned Value (EV) is stochastic** and subject to aggregated contractor performance uncertainty: $EV_i(t) \sim \mathcal{N}(\mu_i(t), \sigma(t)^2)$ (Section 5.1)
- **Revenue recognized is proportional to realized EV**: $\text{Rev}_i(t) = EV_i(t) \times (1 + \rho)$ where $\rho = 0.10$
- **Revenue recognition timing matches EVM measurement cycles** (monthly, Section 3.4)

- **Seasonal variance in EV propagates to revenue recognition**, creating higher revenue uncertainty during year-end periods

Justification:

- Standard practice in IFRS 15 / ASC 606 for construction contracts with continuous transfer of control
- Consistent with EVM-based payment framework where payments are tied to earned value milestones (Section 3.4)
- Maintains tractable relationship between performance and revenue while incorporating realistic uncertainty
- Stochastic EV model (Section 5.1) ensures revenue recognition reflects actual contractor performance variability
- Focus remains on cash timing mismatch (payment delays) while acknowledging performance-driven revenue uncertainty
- Aligns with cashflow-based budgeting assumption (Section 3.3) by using EV as the bridge between cost and revenue

Payment Delays Are Independent and Identically Distributed

- **Each progress payment experiences independent administrative delay** $d_i \sim \text{Uniform}(10, 30)$ days
- **No correlation between delays** across projects or time periods
- **No modeling of client-specific payment behaviors** (e.g., chronic late payers vs. prompt payers)
- **Delay distribution is stationary** and does not change with economic conditions or client financial health

Justification:

- Captures realistic administrative variability without introducing client-level state variables
- Uniform distribution reflects lack of historical data on delay patterns in foundational study
- Independence assumption prevents exponential growth in state space from delay correlations
- Maintains tractable stochastic environment consistent with aggregate uncertainty principle (Section 3.5)
- Sensitivity analysis can test impact of wider delay ranges in future work

Advance Payments Are Immediate and Risk-Free

- **Advance payment deposited at $t = T_i^{\text{start}}$ with zero delay**
- **Advance ratio $\alpha_{\text{adv}} \in \{0.15, 0.20, 0.25\}$ is exogenous** and contractually fixed
- **No modeling of advance payment guarantees**, bank instruments, or collateral requirements
- **Clients always honor advance payment obligations** (no default risk on initial mobilization funds)

Justification:

- Advance payments are contractually secured and processed before project mobilization in EPC practice
- Simplifies initial liquidity modeling without affecting core budget allocation problem
- Default risk on advance payments is negligible in government/major oil company contracts (Section 3.1)
- Allows focus on progress payment delays, which dominate liquidity uncertainty
- Consistent with foundational simplicity: single deterministic advance payment mechanism

Monthly Billing Cycle Alignment

- **Revenue submissions and payment evaluations occur at discrete monthly intervals** aligned with simulation timesteps
- **No intra-month billing** or ad-hoc payment requests
- **All projects follow synchronized billing calendar** regardless of start date
- **Billing cycles align with rolling horizon re-planning frequency** (Section 3.3)

Justification:

- Standard industry practice for EPC contracts (monthly progress reports)
- Aligns with discrete-time MDP formulation and monthly decision epochs (Section 3.3)
- Avoids continuous-time complexity while maintaining realistic cash flow dynamics
- Consistent with EVM measurement cycles (Section 3.4)
- Maintains tractable temporal structure for RL training

3.3 Portfolio Structure Assumptions

Temporal Structure: Rolling Horizon Framework

- **Planning Horizon:** Fixed $H = 12$ periods (months) lookahead window

- **Portfolio Duration:** Variable, determined by $[t_{\text{start}}, t_{\text{end}}]$ where:
 - t_{start} = January of the year containing $\min(\text{project start times})$
 - t_{end} = $\max(\text{project finish times})$
- **Project Lifecycles:** Heterogeneous start dates and durations
- **Re-planning Frequency:** Monthly rolling re-optimization

Justification:

Anchoring t_{start} to the calendar year start of the earliest project ensures explicit capture of **year-end productivity degradation** encoded in the seasonal variance function $\sigma(t)$. Key rationale:

- **Material Year-End Effects:** Contractor performance drops during November–January significantly impact portfolio outcomes. Aligning the temporal origin to January ensures seasonal shocks occur at consistent, interpretable month indices across all episodes.
- **Model Stability:** Mid-year temporal origins introduce shifted or aliased seasonal variance, complicating agent learning. A January anchor stabilizes the annual cycle and enables reliable pattern recognition.
- **Budget Cycle Alignment:** Most organizations operate on January–December fiscal cycles; this alignment improves realism and interpretability.
- **Stochastic Consistency:** Seasonal uncertainties (e.g., $\sigma_{\text{year-end}}$) synchronize cleanly with month indices, eliminating discontinuities or edge cases in the variance structure.
- **Agent Generalization:** Predictable recurrence of high-uncertainty months at fixed positions within episodes enables robust learning of seasonal dynamics.

This design balances realism and tractability without requiring explicit holiday calendars or variable season boundaries.

Handling Variable Horizons:

- Portfolio spanning 26 months: Solve 26 sequential 12-month rolling planning problems
- Projects starting at $t = 7$: Enter planning window when current time $t_{\text{current}} + H \geq 7$
- Projects finishing at $t = 19$: Exit planning window after EV target achieved
- **Project Masking:** Projects not yet started ($t_{\text{current}} < t_{\text{project start}}$) or already completed are masked from the action space and state representation, ensuring the agent only allocates resources to active or imminent projects
- **Termination:** Portfolio concludes when all projects complete or budget exhausts

Myopia Mitigation

- **Terminal Value Function:** Estimates value of projects extending beyond current planning horizon
- **Prevents Short-Termism:** Ensures decisions account for long-term project value

- **Implementation:** Learned as part of RL value function or approximated using remaining BAC

4. Out of Scope Exclusions

For the foundational paper, several modeling elements are intentionally excluded to preserve tractability and maintain a clear focus on the core contribution: **the rolling-horizon reinforcement learning framework for portfolio budgeting decisions.**

Detailed operational mechanisms—such as resource-level scheduling, disaggregated uncertainty sources, multi-objective optimization, heterogeneous contract structures, explicit holiday calendars, and year-end parameter discontinuities are abstracted away in favor of a simplified, portfolio high level representation.

Instead, **many of these effects are absorbed into the seasonal variance component of the aggregated uncertainty parameter.** This allows the RL agent to learn conservative behavior during higher-uncertainty periods while avoiding the need for explicit modeling of each underlying driver.

Each exclusion reflects a deliberate methodological choice. Introducing these elements would significantly increase model dimensionality, calibration requirements, and environmental non-stationarity, potentially obscuring the primary contribution of the study.

At the same time, these components represent **natural extensions of the framework (future work)** and provide clear opportunities for incremental research progression once the foundational RL formulation and experimental results are validated.

Resource-Level Budgeting & Micro-Scheduling

- No modeling of task-level schedules, activity networks, or WBS structures
- No representation of resource calendars, skill categories, or utilization profiles
- No simulation of material lead times or supply-chain task sequences
- No optimization of labor/equipment/material allocation within projects

Justification:

- These require operational project scheduling (CPM/PERT), outside strategic budgeting
- Adds high dimensionality and non-stationarity incompatible with portfolio-level RL
- Focus remains on financial and timing decisions, not execution logistics

Disaggregated Uncertainty

- No separate modeling of human resource variability
- No explicit inflation modeling by category or market segment
- No independent treatment of supply-chain disruption
- No separate uncertainty source for regulatory or policy changes

Justification:

- Q1 models rely on aggregated uncertainty for tractability
- Avoids multi-factor calibration against external economic datasets
- Maintains a stationary stochastic environment needed for stable RL training

Multi-Objective Optimization

- No explicit modeling of strategic value maximization
- No risk-adjusted utility or weighted preference functions
- No stakeholder-specific objective formulations

Justification:

- Q1 problem focuses on single-objective financial optimization
- Multi-objective frameworks introduce Pareto front complexity
- Requires stakeholder preference modeling outside the scope of automated RL

Advanced EVM Variations

- No mixed or hybrid payment schemes (EVM + milestones)
- No portfolio with multiple payment logics simultaneously
- No incentive/penalty mechanisms tied to performance metrics

Justification:

- Single EVM model ensures uniform performance and payment measurement
- Simplifies reward computation and state transitions
- Avoids heterogeneous contract structures that complicate the RL environment

Explicit Holiday Calendars

- No country-specific holiday calendars or productivity shutdown periods
- No modeling of seasonal workforce productivity

- No date-specific scheduling impacts

Justification:

- Calendar-driven productivity patterns introduce fine-grained temporal non-stationarity
- Requires region-specific datasets outside Q1 scope
- Q1 uses continuous time approximations, not calendar simulation

Year-End Parameter Discontinuities

- No modeling of price index resets at new-year boundaries
- No explicit regulatory update cycles at year-end
- No contract renegotiation events tied to fiscal closure
- No fiscal-year budget resets or rollover constraints

Justification:

- Year-end effects introduce deterministic discontinuities that break stationarity
- Requires modeling economic policy cycles and real fiscal data calibration
- Necessitates separating deterministic vs. stochastic drivers
- Complicates reward design by adding discontinuous incentive structures

Excluded: Project-Specific S-Curve Calibration

- No estimation of **individual project spending profiles** from historical data
- No modeling of **phase-specific spending patterns** (e.g., 20% engineering, 30% procurement, 45% construction)
- No representation of **project type heterogeneity** (onshore vs. offshore, greenfield vs. brownfield)
- No dynamic adjustment of S-curve shape based on **real-time project performance**

Reason for Exclusion:

- Requires detailed historical cashflow data rarely available at portfolio planning stage
- Introduces $2N$ additional parameters inconsistent with foundational simplicity principle (Section 3.2)
- Phase-level modeling belongs to operational execution layer, not strategic budgeting (Section 4)
- Performance-based adjustments require real-time monitoring outside Q1 scope
- Aggregate uncertainty model (Section 3.5) absorbs project-level variability

Excluded: Variable Profit Margins and Competitive Pricing

- No modeling of **bid competitiveness** or market-driven pricing
- No representation of **risk-adjusted margins** (higher profit for higher uncertainty projects)
- No **client negotiation dynamics** or discount structures
- No **incentive fee structures** (e.g., cost-plus-incentive-fee contracts)

Reason for Exclusion:

- Government contracts with fixed 10% ROI eliminate pricing competition in target domain
- Variable margins introduce contract-level uncertainty orthogonal to budget allocation problem
- Competitive bidding requires game-theoretic modeling outside RL portfolio optimization scope
- Maintains single contractual framework consistent with foundational approach (Section 3.2)
- Focus is on managing fixed-margin contracts under payment uncertainty, not winning variable-margin bids

Excluded: Milestone-Based and Output-Based Revenue Recognition

- No modeling of **discrete milestone payments** (e.g., 30% at design approval, 60% at mechanical completion)
- No representation of **deliverable-based billing** (per equipment unit, per facility module)
- No **retainage/holdback provisions** (e.g., 10% withheld until final acceptance)
- No **performance bonuses** or liquidated damages affecting revenue magnitude or timing

Reason for Exclusion:

- Milestone structures create discontinuous revenue jumps incompatible with continuous EV-based progress measurement (Section 3.4)
- Output-based billing requires detailed scope decomposition outside strategic budgeting level (Section 4)
- Retainage modeling adds state complexity (tracking withheld amounts per project) without altering core allocation problem
- Performance incentives beyond the fixed 10% margin introduce additional outcome uncertainty better addressed in future multi-objective extensions (Section 4)
- Maintains consistency with EVM-based continuous progress measurement where revenue follows realized EV with stochastic variance (Section 5.1)
- The model already captures performance-driven revenue variability through stochastic EV; milestone-based recognition would introduce orthogonal structural complexity

Excluded: Client-Specific Payment Behavior and Default Risk

- No modeling of **client creditworthiness** or financial distress
- No representation of **systematic payment delays** by specific clients (e.g., government agencies with budget cycles)
- No **default risk** or **partial payment scenarios**
- No **currency risk** or **cross-border payment complications**

Reason for Exclusion:

- Client-level heterogeneity requires tracking payment history and credit ratings per client
- Default risk in major oil & gas EPC contracts is negligible (clients are typically investment-grade entities, Section 3.1)
- Systematic delays (e.g., fiscal year-end bottlenecks) are absorbed into aggregate seasonal variance (Section 3.5)
- Currency and cross-border issues belong to financial risk management, not operational budgeting
- Maintains stationary stochastic environment required for stable RL training (Section 3.5)

Excluded: Intra-Month Cash Flow Dynamics

- No modeling of **daily or weekly cash positions**
- No representation of **short-term borrowing** or **overdraft facilities** to cover intra-month gaps
- No **cash flow forecasting** at sub-monthly granularity
- No **working capital optimization** within billing cycles

Reason for Exclusion:

- Monthly timesteps align with strategic planning horizon and billing cycles (Section 3.3)
- Intra-month liquidity management is a treasury function, not portfolio allocation decision
- Short-term financing instruments (credit lines, overdrafts) are operational tools outside RL action space
- Sub-monthly granularity increases computational cost without improving allocation policy quality
- Consistent with cashflow-based budgeting at strategic level (Section 3.3)

Excluded: Billing Disputes and Payment Holds

- No modeling of **invoice rejections** due to scope disagreements
- No representation of **rework-related payment suspensions**
- No **change order negotiations** affecting payment timing
- No **lien/claim processes** or legal disputes delaying payments

Reason for Exclusion:

- Billing disputes are rare in well-managed EPC projects with clear scope definitions
- Rework and change orders introduce project execution risks outside financial planning scope (Section 4)
- Legal disputes are low-probability events that do not affect baseline portfolio strategy
- Focus is on normal payment variability (administrative delays), not exceptional disruptions
- Maintains tractable stochastic model consistent with aggregate uncertainty principle (Section 3.5)

5. Modeling Scope & Simplifications

⚠ needs checking (inflow outflow modeling)

3.2 Project Cashflow Representation

Each project i is characterized by a **cumulative spending S-curve** $S_i(t)$, representing the contract planned fraction of Budget at Completion (BAC) spent by time t within the project duration:

$$C_i(t) = \text{Cumulative planned spend of project } i \text{ at time } t$$
$$S_i(t) = \frac{C_i(t)}{\text{BAC}_i}, \quad t \in [T_i^{\text{start}}, T_i^{\text{end}}]$$

where:

- BAC_i = Budget at Completion for project i
- T_i^{start} = project start time
- T_i^{end} = project completion time
- $D_i = T_i^{\text{end}} - T_i^{\text{start}}$ = project duration
- $S_i(t) \in [0, 1]$
- $C_i(t) = S_i(t) \times \text{BAC}_i$

S-curve Parameterization for EPC Projects

Following empirical studies of oil & gas EPC projects, we model $S_i(t)$ using a **Beta cumulative distribution function (Beta CDF)**:

$$S_i(\tau) = I_\tau(\alpha, \beta) = \frac{\int_0^\tau u^{\alpha-1}(1-u)^{\beta-1}du}{B(\alpha, \beta)}$$

where:

- $\tau = \frac{t - T_i^{\text{start}}}{D_i} \in [0, 1]$ = normalized project progress
- $\alpha, \beta > 0$ = shape parameters
- $B(\alpha, \beta) = \int_0^1 u^{\alpha-1}(1-u)^{\beta-1}du$ = Beta function (normalization constant)

Physical Interpretation:

- α controls front-loading: higher $\alpha \rightarrow$ more spending early
- β controls back-loading: higher $\beta \rightarrow$ more spending late
- $\alpha = \beta \rightarrow$ symmetric S-curve
- $\alpha > \beta \rightarrow$ front-loaded (typical for EPC projects)

Literature-Based Calibration

Empirical Evidence from EPC Projects:

1. **Barraza & Bueno (2007)** — "Modeling the S-curve for construction projects"
 - Analyzed 23 industrial construction projects
 - Found $\alpha \in [2.5, 3.0]$, $\beta \in [2.0, 2.5]$ for typical EPC workflows
 - Peak spending rate occurs at 40-45% completion
2. **Cioffi (2005)** — "A tool for managing projects: an analytic parameterization of the S-curve"
 - Demonstrated Beta distribution provides superior fit to construction cashflows vs. polynomial models
 - Identified α/β ratio as key determinant of front-loading intensity
3. **Miskawi (1989)** — "An S-curve equation for project control"
 - Oil & gas projects exhibit moderate front-loading due to engineering-procurement-construction sequencing
 - Engineering phase (15-20% of budget) drives early spending acceleration
4. **Kenley & Wilson (1986)** — "A construction project cash flow model"
 - Validated S-curve shape consistency across project sizes within same sector

- Portfolio-level aggregation smooths individual project variations

Our Baseline Parameters:

$$\alpha = 2.5, \quad \beta = 2.0$$

This yields:

- **Peak spending rate** at $\tau^* = \frac{\alpha-1}{\alpha+\beta-2} = \frac{1.5}{2.5} = 0.60$ (60% completion)
- **Inflection point** (maximum acceleration) at $\tau_{\text{inflection}} \approx 0.43$ (43% completion)
- **Moderate front-loading** consistent with EPC workflow phases:
 - Engineering (15-20%): Design, specifications, procurement planning
 - Procurement (25-30%): Equipment ordering, long-lead items
 - Construction (40-50%): Field installation, civil works
 - Commissioning (5-10%): Testing, startup, handover

Calibration Validation:

- At $\tau = 0.25$: $S(0.25) \approx 0.16$ (16% spent at 25% timeline) — matches engineering phase completion
- At $\tau = 0.50$: $S(0.50) \approx 0.50$ (50% spent at 50% timeline) — symmetric midpoint
- At $\tau = 0.75$: $S(0.75) \approx 0.84$ (84% spent at 75% timeline) — construction phase dominance

Rationale for Uniform S-curve Across Projects

Why we don't model project-specific shapes:

1. **State space explosion:** Allowing individual (α_i, β_i) per project adds $2N$ state dimensions without strategic value at portfolio planning level
2. **Portfolio-level robustness:** Budget allocation decisions depend on aggregate cashflow dynamics, which smooth out individual project variations through the law of large numbers
3. **Data limitations:** Estimating project-specific S-curve parameters requires detailed historical tracking (weekly/monthly cashflow records) rarely available at strategic planning stage
4. **Empirical clustering:** Literature shows EPC projects within same sector exhibit similar spending profiles (Barraza & Bueno 2007; Kenley & Wilson 1986)
5. **Sensitivity analysis validation:** Optimal policies remain stable across reasonable parameter ranges:
 - $\alpha \in [2.0, 3.0]$ (light to moderate front-loading)
 - $\beta \in [1.5, 2.5]$ (symmetric to moderate back-loading)

Mathematical Properties

Incremental spending rate (cashflow velocity) at time t :

$$\frac{dC_i}{dt} = \frac{BAC_i}{D_i} \cdot \frac{dS_i}{d\tau} = \frac{BAC_i}{D_i} \cdot \text{Beta-PDF}(\tau; \alpha, \beta)$$

where the Beta probability density function is:

$$\text{Beta-PDF}(\tau; \alpha, \beta) = \frac{\tau^{\alpha-1}(1-\tau)^{\beta-1}}{B(\alpha, \beta)}$$

Peak spending rate occurs at:

$$\tau^* = \frac{\alpha - 1}{\alpha + \beta - 2} = 0.60 \quad (\text{for } \alpha = 2.5, \beta = 2.0)$$

Cumulative spending at key milestones:

- 25% timeline: $S(0.25) = I_{0.25}(2.5, 2.0) \approx 0.16$ (16% budget consumed)
- 50% timeline: $S(0.50) = I_{0.50}(2.5, 2.0) \approx 0.50$ (50% budget consumed)
- 75% timeline: $S(0.75) = I_{0.75}(2.5, 2.0) \approx 0.84$ (84% budget consumed)

5.3 Project Revenue Inflow Model

Overview

Project revenue follows a **Cost-Plus pricing structure** with 10% ROI markup, reflecting government value-added regulations. Revenue recognition follows the **Percentage-of-Completion(POC) method** (IFRS 15 / ASC 606) based on stochastic Earned Value, while cash inflow follows a two-phase payment structure with administrative delays.

Contract Pricing Structure

For project i with Budget at Completion BAC_i :

$$\text{Contract Value}_i = BAC_i \times (1 + \rho)$$

where:

- $\rho = 0.10$ = target Return on Investment (ROI) / profit margin
- $\text{Gross Profit}_i = \text{BAC}_i \times \rho$

Rationale:

- Cost-Plus contracts are standard in oil & gas EPC sector for large-scale projects with scope uncertainty (Bower 2003)
- Fixed 10% margin reflects government value-added regulations for public sector contracts
- Eliminates pricing competition to focus on operational execution and liquidity management

Revenue Recognition: Percentage-of-Completion with Stochastic EV

Planned Earned Value (EV) — baseline progress measurement:

$$\mu_i(t) = \text{BAC}_i \cdot I_\tau(\alpha = 2.5, \beta = 2.0)$$

where $\tau = \frac{t - T_i^{\text{start}}}{D_i}$ is normalized project progress.

Realized Earned Value — actual performance with uncertainty:

$$EV_i(t) \sim \mathcal{N}(\mu_i(t), \sigma(t)^2)$$

where:

- $\mu_i(t)$ = planned EV following cost S-curve shape
- $\sigma(t)$ = time-varying performance uncertainty (Section 5.1)
- $\sigma(t)$ increases during year-end periods (November-January) to reflect seasonal productivity slowdowns

Recognized Revenue — accounting revenue at time t :

$$\text{Rev}_i(t) = EV_i(t) \times (1 + \rho)$$

Components:

- Cost recovery: $EV_i(t)$ (stochastic, reflects actual work performed)
- Profit recognition: $EV_i(t) \times \rho$ (proportional to earned value)

Literature Support:

- **IFRS 15 / ASC 606:** Revenue recognition for construction contracts requires measurement of progress toward completion

- **Fleming & Koppelman (2016)** — "Earned Value Project Management": EV-based revenue recognition aligns financial reporting with physical progress
- **Christensen (1998)** — "The costs and benefits of the earned value management process": EV provides unbiased estimator of project completion percentage when properly measured

Payment Structure

Phase 1: Advance Payment

- **Timing:** Upon project initiation (period $t = T_i^{\text{start}}$)
- **Amount:** $R_{\text{adv},i} = \alpha_{\text{adv}} \cdot \text{Contract Value}_i$ where $\alpha_{\text{adv}} \in [0.15, 0.25]$
- **Characteristics:**
 - Immediate deposit (no delay)
 - One-time payment
 - Includes proportional profit: $\alpha_{\text{adv}} \times \text{BAC}_i \times \rho$
 - Reduces client financial risk and provides initial working capital

Literature Calibration:

- **Ling et al. (2005)** — "Key project success factors in international construction joint ventures": Advance payments in oil & gas EPC contracts typically range 15-25% of contract value
- **Kangari & Bakheet (2001)** — "Construction surety bonding": Advance payments serve as mobilization funds for equipment procurement and site setup

Example: If $\text{BAC}_i = \$1M$, $\rho = 0.10$, $\alpha_{\text{adv}} = 0.20$:

$$R_{\text{adv},i} = 0.20 \times \$1.1M = \$220K \quad (\$200K \text{ cost recovery} + \$20K \text{ profit})$$

Phase 2: Progress-Based Payments

- **Trigger:** Activated after recognized revenue exceeds advance payment: $\text{Rev}_i(t) \geq R_{\text{adv},i}$
- **Submission Cycle:** Monthly evaluation at period t aligned with EVM measurement cycles
- **Payment Calculation:**

$$R_i(t) = \text{Rev}_i(t) - \sum_{\tau=T_i^{\text{start}}}^{t-1} R_i(\tau)$$

(Revenue increment since last payment, based on realized EV)

Administrative Delay

- **Evaluation & Approval:** $d_{\text{eval},i} \sim \text{Uniform}(10, 30)$ days
- **Actual Deposit:** Revenue $R_i(t)$ submitted at period t is deposited at period $t + \lceil d_{\text{eval},i}/30 \rceil$
- **Stochastic Component:** Delay varies by project and submission, reflecting:
 - Client review processes (invoice verification, EV audit)
 - Documentation completeness (progress reports, photos, test results)
 - Approval workflows (multi-level sign-offs in large organizations)
 - Banking transfer times (international wire transfers, currency conversion)

Literature Calibration:

- **Odeh & Battaineh (2002)** — "Causes of construction delay: traditional contracts": Payment delays in Middle East construction projects average 20-45 days
- **Assaf & Al-Hejji (2006)** — "Causes of delay in large construction projects": Administrative processing accounts for 15-30 day delays in 70% of projects
- **Uniform(10, 30) distribution:** Conservative estimate capturing typical range without modeling extreme outliers (disputes, defaults)

Cash Inflow Function

For project i at timestep t :

$$\text{CashIn}_i(t) = \begin{cases} \alpha_{\text{adv}} \cdot \text{BAC}_i \times (1 + \rho) & \text{if } t = T_i^{\text{start}} \\ \max \left(0, \text{Rev}_i(t - d_i) - \sum_{\tau=T_i^{\text{start}}}^{t-1} \text{CashIn}_i(\tau) \right) & \text{if } t > T_i^{\text{start}} + d_i \\ 0 & \text{otherwise} \end{cases}$$

where:

- $d_i = \lceil d_{\text{eval},i}/30 \rceil$ = delay in periods for project i 's current submission
- $\text{Rev}_i(t - d_i)$ = recognized revenue at delayed time (based on realized EV)
- $\max(0, \cdot)$ ensures non-negative cash inflow (no overpayment recovery)

Key Distinction:

- $\text{Rev}_i(t)$ = accounting revenue (stochastic, based on realized EV)

- $\text{CashIn}_i(t)$ = actual cash received (stochastic + delayed)

Portfolio-Level Dynamics

Total Cash Inflow at period t :

$$B_{\text{inflow}}(t) = \sum_{i=1}^n \text{CashIn}_i(t)$$

Budget Update:

$$B_{\text{available}}(t) = B_{\text{rem}}(t-1) - \sum_i C_i(t-1) + B_{\text{inflow}}(t)$$

Cumulative Profit (accounting, not cash):

$$\Pi(t) = \sum_{i=1}^n [\text{Rev}_i(t) - C_i(t)] = \sum_{i=1}^n \text{BAC}_i \times \rho \cdot \frac{EV_i(t)}{\text{BAC}_i}$$

Simplifying:

$$\Pi(t) = \rho \sum_{i=1}^n EV_i(t)$$

Stochastic Profit: Since $EV_i(t)$ is stochastic, cumulative profit is also stochastic even with fixed margin ρ .

Model Parameters

Parameter	Distribution	Value/Range	Literature Source
Profit margin (ρ)	Fixed	0.10	Government value-added regulation (10% ROI)
Advance payment ratio (α_{adv})	Uniform	[0.15, 0.25]	Ling et al. (2005), Kangari & Bakheet (2001)
Evaluation delay ($d_{\text{eval},i}$)	Uniform	[10, 30] days	Odeh & Battaineh (2002), Assaf & Al-Hejji (2006)

Parameter	Distribution	Value/Range	Literature Source
Submission frequency	Deterministic	Monthly	Standard EVM practice (Fleming & Koppelman 2016)
EV uncertainty ($\sigma(t)$)	Time-varying	Section 5.1	Seasonal productivity variance

Critical Differences: Cost vs Revenue Dynamics

Aspect	Cost Outflow	Revenue Recognition	Cash Inflow
Magnitude	BAC_i	$BAC_i \times 1.10$	$BAC_i \times 1.10$
Shape (baseline)	$I_\tau(2.5, 2.0)$	$I_\tau(2.5, 2.0)$	$I_\tau(2.5, 2.0)$
Stochasticity	Deterministic (planned)	Stochastic (via $EV_i(t)$)	Stochastic + Delayed
Timing	Immediate (as work progresses)	Immediate (as EV realized)	Delayed by $d_i \sim U(10, 30)$ days
Purpose	Cash outflow planning	Accounting compliance	Liquidity management
Accounting	Cost recognition = Cash outflow	Revenue $\neq$ Cash inflow	Cash $\neq$ Revenue

Key Insight: Revenue S-curve follows the same **shape** as cost S-curve under baseline conditions, but:

1. **Scaled by** $(1 + \rho)$ to include profit margin
2. **Stochastic** due to performance uncertainty in realized EV
3. **Time-shifted** for cash inflow due to administrative delays

The combination of stochastic EV and payment delays creates **dual uncertainty** in liquidity forecasting that the RL agent must manage.

Agent Planning Impact

State Augmentation:

- **Advance payment status:** $1_{t \geq T_i^{\text{start}}}$ (binary indicator)
- **Pending revenue:** $\sum_{\tau=t-d_{\max}}^{t-1} R_i(\tau)$ for $\tau \in [t - d_{\max}, t]$ (in transit, not yet deposited)
- **Expected inflow:** $\mathbb{E}[B_{\text{inflow}}(t+1) | EV_i(t)]$ based on current realized EV trajectories
- **Revenue variance:** $\text{Var}[B_{\text{inflow}}(t+1)]$ to quantify liquidity risk

Liquidity Constraint:

The agent must anticipate both performance uncertainty and payment delays when planning allocations. Even with positive expected cumulative profit $\mathbb{E}[\Pi(t)] > 0$, the portfolio may face cash shortages if:

$$B_{\text{available}}(t) < \sum_i C_i(t)$$

This necessitates:

1. **Strategic timing** of project starts to maintain positive cash flow
2. **Buffer management** to absorb EV variance and payment delay shocks
3. **Risk-aware allocation** that accounts for seasonal variance spikes (year-end periods)

Reward Function Implication:

The reward must balance:

- **Throughput maximization:** Complete high-value projects quickly
- **Liquidity preservation:** Avoid bankruptcy from cash flow timing mismatches
- **Profit realization:** Ensure sufficient EV progress to trigger revenue recognition

This creates a multi-objective optimization problem where the RL agent learns to trade off immediate project progress against future liquidity risk.

5.1 Aggregated Contractor Performance Uncertainty with Seasonal Variation

To model contractor behavior in a tractable yet realistic manner, all sources of performance variability are consolidated into a single stochastic uncertainty parameter. This parameter governs deviations between planned progress and realized Earned Value (EV). Seasonal productivity effects, including year-end holiday slowdowns, are incorporated through a time-dependent variance structure.

5.1.1 Unified Stochastic Representation

For each project i and time period t , realized Earned Value is modeled as a normally distributed random variable:

$$EV_i(t) \sim \mathcal{N}(\mu_i(t), \sigma(t)^2)$$

where:

- $\mu_i(t)$ is the planned progress for project i ,
- $\sigma(t)$ captures the aggregated performance uncertainty.

The unified uncertainty parameter $\sigma(t)$ represents the combined effects of:

- workforce availability and performance fluctuations
- inflation and macroeconomic conditions
- supply chain instability
- short-term regulatory shifts
- technical and engineering challenges
- **seasonal productivity variation**

These factors are not modeled individually; instead, their joint effect is absorbed into the stochastic variance term.

5.1.2 Seasonal Variance Structure

Contractor productivity typically declines around the year-end period (November, December, January). Instead of encoding explicit holiday calendars, this effect is introduced by allowing the uncertainty parameter to vary seasonally.

Let the model operate in monthly time steps with $t \bmod 12$ denoting the month of the year (0 = December, 1 = January, ..., 11 = November). The variance is defined piecewise:

$$\sigma(t) = \begin{cases} \sigma_{\text{baseline}}, & t \bmod 12 \in \{2, 3, \dots, 10\} \\ \sigma_{\text{year-end}}, & t \bmod 12 \in \{11, 0, 1\} \end{cases}$$

with:

$$\sigma_{\text{year-end}} \approx 1.5 \sigma_{\text{baseline}}$$

indicating a 50% increase in performance variance during the high-uncertainty season. This elevates the risk of underperformance near the year-end and encourages the RL agent to adopt more

conservative allocation strategies during these periods.

5.1.3 Modeling Rationale and Validation

This formulation is intentionally designed as an **aggregated contractor performance uncertainty model with seasonal variation**. It provides:

- a compact, low-dimensional representation suitable for RL training
- sufficient realism to induce seasonally-aware agent behavior
- a controlled structure that avoids explicit holiday calendars and discontinuities

Validation and sensitivity analysis will evaluate how different choices of σ_{baseline} and $\sigma_{\text{year-end}}$ affect system behavior. This approach is documented as a modeling simplification, with future extensions enabling explicit modeling of calendar effects or disaggregated uncertainty sources.

5.2 Rolling Horizon Framework Details

Planning Window Structure

At each timestep t :

- **Observation Window**: Current portfolio state
- **Planning Horizon**: $[t, t + H]$ where $H = 12$ periods
- **Decision**: Complete budget allocation plan for entire horizon $[t, t + H]$
- **Execution**: Only the allocation for period t is executed; plan for $[t + 1, t + H]$ guides current decision but remains uncommitted

State Representation

For each project i at timestep t :

Project-level state (only for unmasked projects):

- Progress ratio: $p_i = \frac{EV_i(t)}{BAC_i}$
- Remaining duration: $r_i = \text{finish}_i - t$
- Performance trend: SPI_i (moving average over last 3 periods)
- Active status: $\alpha_i \in \{0, 1\}$
- Time to start: $s_i = \max(0, \text{start}_i - t)$

Portfolio-level state:

- Remaining budget: $B_{\text{rem}}(t)$

- Current period: t
- Periods to portfolio end: $t_{\text{end}} - t$
- Active project count: $n_{\text{active}}(t)$
- Seasonal indicator: $(t - t_{\text{start}}) \bmod 12$ (captures year-end effects relative to portfolio origin)

Project Masking:

- Projects with $t < \text{start}_i$ (not yet started) or $EV_i(t) = BAC_i$ (completed) are masked from state and action space
- Masked projects contribute zero to state vectors and receive zero allocation

Action Space

- **Full horizon budget plan:** $\mathbf{B}(t) = [\mathbf{b}(t), \mathbf{b}(t+1), \dots, \mathbf{b}(t+H)]$
 - Where $\mathbf{b}(\tau) = [b_1(\tau), b_2(\tau), \dots, b_n(\tau)]$ for each period $\tau \in [t, t+H]$
- **Execution:** Only $\mathbf{b}(t)$ is applied to the environment
- **Constraints:**
 - Period-wise budget: $\sum_i b_i(\tau) \leq B_{\text{available}}(\tau)$ for all $\tau \in [t, t+H]$
 - Cumulative budget: $\sum_{\tau=t}^{t+H} \sum_i b_i(\tau) \leq B_{\text{rem}}(t)$
 - Only unmasked, active projects receive non-zero allocations
 - Minimum payment: $b_i(\tau) \geq b_{\min,i}$ if project i is active at τ

Episode Termination

An episode ends when:

1. $t = t_{\text{end}}$ (all projects completed or reached latest finish time), OR
2. Budget exhausted ($B_{\text{rem}}(t) = 0$), OR
3. All projects masked (none active or imminent within horizon)

Reward Function

Immediate reward at timestep t :

$$r(t) = \sum_i [EV_i(t) - EV_i(t-1)] - \text{penalty}_{\text{budget overrun}} - \text{penalty}_{\text{project delay}}$$

Terminal reward (at episode end):

$$r_{\text{terminal}} = \sum_i EV_i(t_{\text{end}}) + \text{terminal_value}(\text{incomplete projects})$$

where `terminal_value` estimates remaining project value for any unfinished work beyond t_{end}

Handling Edge-Case Portfolio Scenarios

The framework is designed to handle diverse portfolio configurations through its masking and temporal anchoring mechanisms:

Scenario 1: Projects start after portfolio origin

Example: Portfolio spans January–December, but all projects start in March

- **Handling:** Projects are masked for $t \in [\text{Jan}, \text{Feb}]$
- **Behavior:** Agent produces horizon plan with zero allocations for masked projects; no execution during gap months
- **Impact:** Episode progresses normally; seasonal indicator still tracks from January origin

Scenario 2: Temporal gaps between projects

Example: Project A finishes in April, Project B starts in July

- **Handling:** Both projects masked during $t \in [\text{May}, \text{Jun}]$
- **Behavior:** Agent plans ahead for Project B activation; no execution during gap months
- **Impact:** Budget preserved; episode continues until t_{end}

Scenario 3: Planning starts mid-project

Example: Planning begins in June for projects that started in February

- **Handling:** Historical data provided as initial state at $t_{\text{planning_start}}$
 - Client supplies: $EV_i(t)$, $AC_i(t)$, $SPI_i(t)$, $CPI_i(t)$ for all $t < t_{\text{planning_start}}$
 - These metrics become the **initial conditions** for agent decision-making
- **Behavior:**
 - Agent receives actual performance history (Feb–May) as part of state at $t = \text{June}$
 - Produces first horizon plan at $t_{\text{planning_start}}$ using real historical context
 - Executes $\mathbf{b}(\text{June})$ from that plan
- **Implementation:** Introduce parameter $t_{\text{planning_start}} \geq t_{\text{start}}$
 - For $t < t_{\text{planning_start}}$: historical data only (no agent actions)
 - For $t \geq t_{\text{planning_start}}$: agent actively plans and executes
- **Impact:** Framework supports both fresh portfolio starts ($t_{\text{planning_start}} = t_{\text{start}}$) and mid-project entry with known history

Design Rationale:

- **Temporal origin** (t_{start}) remains anchored to January for seasonal consistency and model generalization

- **Planning start** ($t_{\text{planning_start}}$) decouples decision-making from temporal origin, enabling flexible deployment
- **Horizon planning** with single-period execution balances long-term foresight with adaptive replanning as uncertainty resolves
- **Masking** ensures agent never acts on unavailable or completed projects, maintaining action space validity across all scenarios

6. Future Work & Extensions

This section outlines natural extensions of the foundational methodology established in this work. Each direction represents a scientifically rigorous path for follow-up research.

6.1 Portfolio Composition Specialization: Local vs. International Projects

Research Question

Do portfolios dominated by domestic vs. international projects require specialized model variants, or does a unified pre-trained model generalize across composition ratios?

Motivation

The current work employs a **unified model** pre-trained on international EPC literature and fine-tuned on company-specific data. However, real-world portfolios exhibit varying compositions:

- **Domestic-heavy portfolios:** Government infrastructure, national energy projects
- **International-heavy portfolios:** Multinational EPC contractors, cross-border joint ventures
- **Mixed portfolios:** Diversified project companies balancing local and international work

These portfolio types may exhibit distinct operational characteristics:

Characteristic	Domestic Projects	International Projects
Payment delays	1-2 periods (local banking)	3-5 periods (wire transfers, currency conversion)
Seasonal volatility	High year-end spikes (fiscal cycles)	Smoother (diversified calendar effects)

Characteristic	Domestic Projects	International Projects
Regulatory constraints	Strict local compliance deadlines	Variable cross-border coordination
Contractor behavior	Known local performance patterns	Higher uncertainty from unfamiliar vendors

Proposed Extension: Three Model Variants

Variant 1: Fully Local Model

- **Specialization:** Domestic-only project portfolios
- **Fine-tuning protocol:** Country-specific contractor performance distributions, local regulatory payment schedules, domestic banking delays
- **Use case:** National infrastructure agencies, domestic energy companies

Variant 2: Optimal Mixed Model

- **Specialization:** Balanced local + international portfolio compositions
- **Sensitivity analysis:** Performance across local/international weight ratios (100%/0%, 75%/25%, 50%/50%, 25%/75%, 0%/100%)
- **Use case:** Diversified EPC contractors, multinational project portfolios

Variant 3: Fully International Model

- **Specialization:** International-only project portfolios
- **Fine-tuning protocol:** Cross-border payment delays, currency risk patterns, international contracting norms
- **Use case:** Global EPC firms, offshore energy projects

Key Research Challenges

1. Data Availability

- Requires empirical calibration of **country-specific contractor performance distributions**
- Current literature lacks granular data on domestic vs. international cashflow volatility patterns
- Synthetic data generation must be validated against regional industry benchmarks

2. Model Architecture Decision

- **Option A:** Train 3 separate models (higher specialization, 3× training cost)

- **Option B:** Single model with portfolio composition as input feature (lower cost, potential generalization loss)
- **Trade-off analysis:** Specialization benefit vs. computational burden

3. Sensitivity Analysis Design

- Define threshold ratios where fine-tuning becomes necessary (e.g., >80% local projects)
- Measure performance degradation as portfolio composition deviates from training distribution
- Establish guidelines: "When should practitioners use Variant 1 vs. Variant 2 vs. Variant 3?"

Experimental Design (Lightweight Approach)

To address this question **without full model re-architecture**, a sensitivity analysis can be conducted:

1. **Generate portfolios** with varying local/international ratios: 100%/0%, 75%/25%, 50%/50%, 25%/75%, 0%/100%
2. **Define project types** by operational parameters:
 - **Local projects:** Payment delay $\sim$ Uniform(1, 2 periods), seasonal volatility $\sigma_{\text{year-end}} = 0.25$
 - **International projects:** Payment delay $\sim$ Uniform(3, 5 periods), seasonal volatility $\sigma_{\text{year-end}} = 0.15$
3. **Train one unified model** on mixed portfolios (50%/50% baseline)
4. **Test performance** across all composition ratios
5. **Analyze:**
 - Plot: Portfolio composition (x-axis) vs. NPV/Budget utilization (y-axis)
 - Identify: Composition ranges where performance degrades >10%
 - Conclude: When fine-tuning for specific compositions becomes necessary

Expected Outcome: Either (a) unified model generalizes well across all compositions, or (b) performance degrades at extremes (>80% local or >80% international), justifying specialized variants.

Scientific Contribution

- **Methodological:** Establishes when portfolio composition warrants model specialization vs. unified approach
- **Practical:** Provides decision framework for practitioners selecting model variants
- **Theoretical:** Extends transfer learning theory to portfolio composition heterogeneity

Timeline & Scope

- **Effort:** 2-3 weeks (1 week data generation, 1 week training/testing, 1 week analysis)
- **Positioning:** Natural follow-up paper (Q2/Q3 journal) or extended journal version

- **Prerequisite:** Foundational Q1 paper must establish baseline unified model first